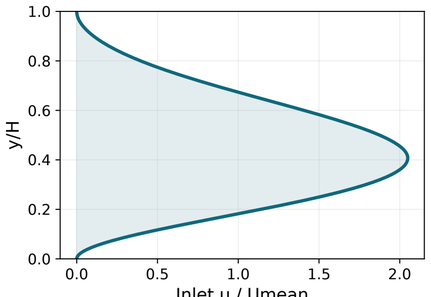
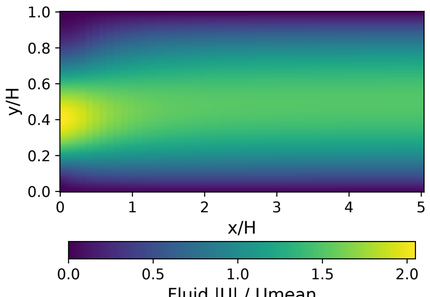
PI-CFO | Asymmetric inlet | Re=30 | velocity L2: 1.603% | pressure L2: 2.374%

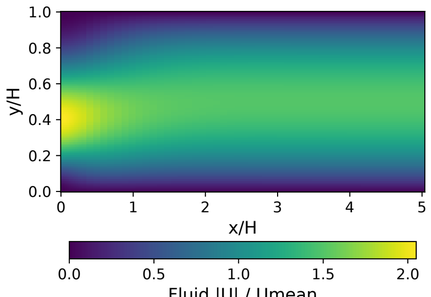
01 Input inlet profile



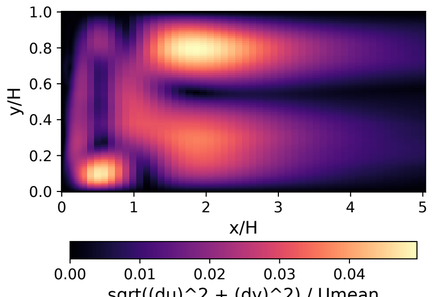
02 PI-CFO prediction



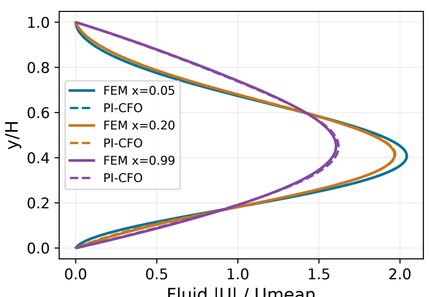
03 Independent FEM



04 Absolute vector error



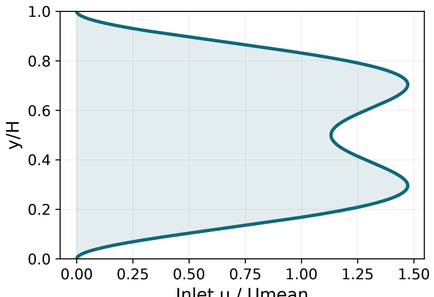
05 Cross-section check



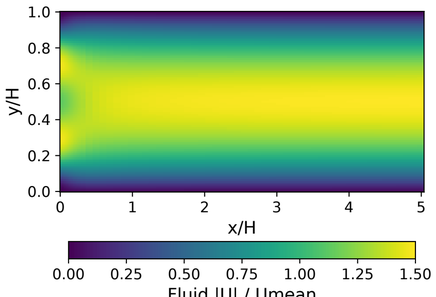
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

PI-CFO | Twin-peaked inlet | Re=20 | velocity L2: 0.946% | pressure L2: 1.833%

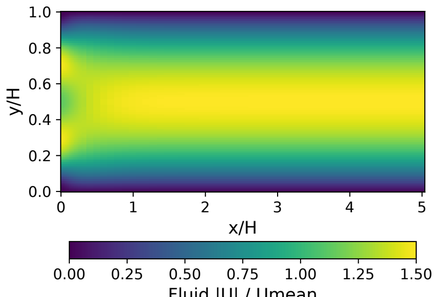
01 Input inlet profile



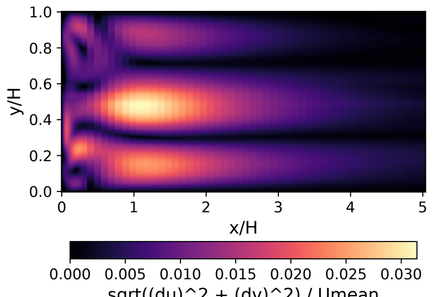
02 PI-CFO prediction



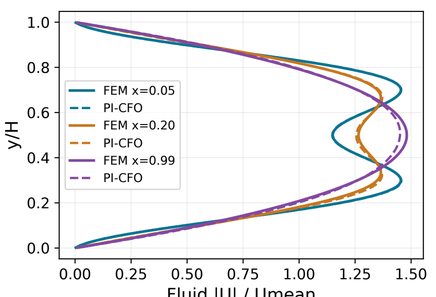
03 Independent FEM



04 Absolute vector error



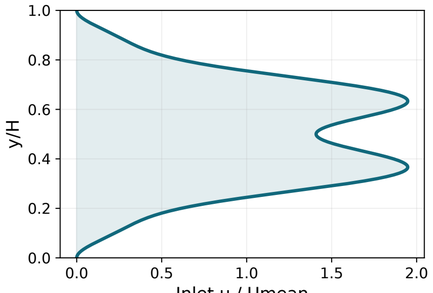
05 Cross-section check



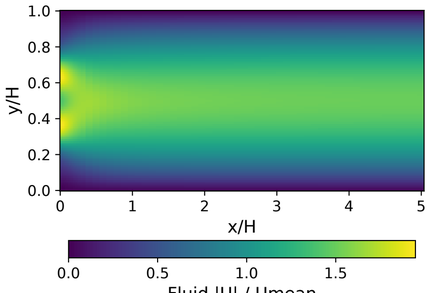
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

PI-CFO | Wavy inlet | Re=20 | velocity L2: 1.801% | pressure L2: 5.043%

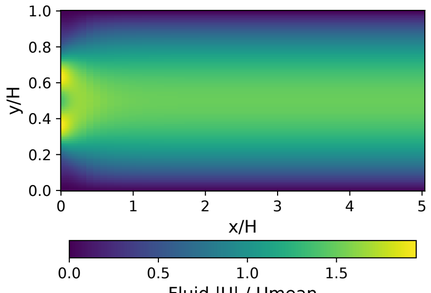
01 Input inlet profile



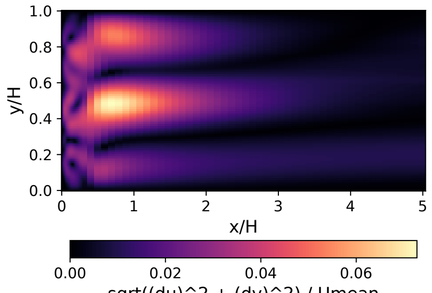
02 PI-CFO prediction



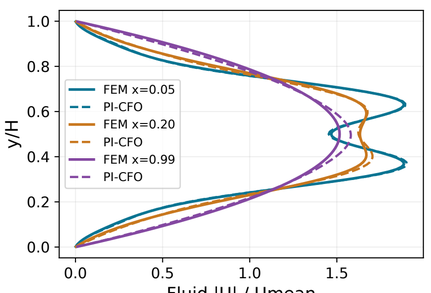
03 Independent FEM



04 Absolute vector error

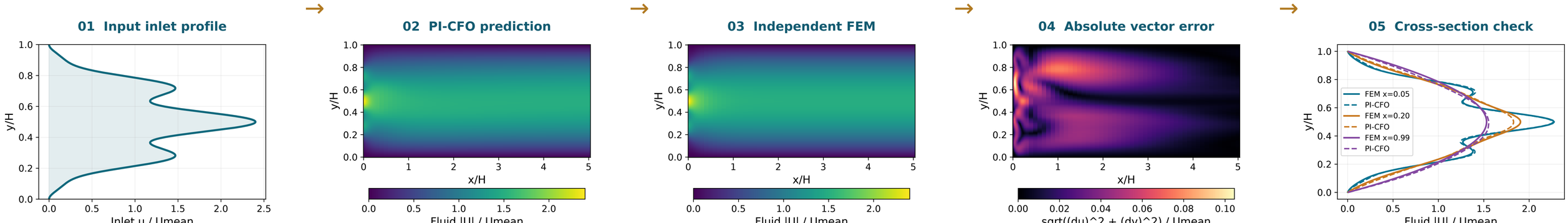


05 Cross-section check



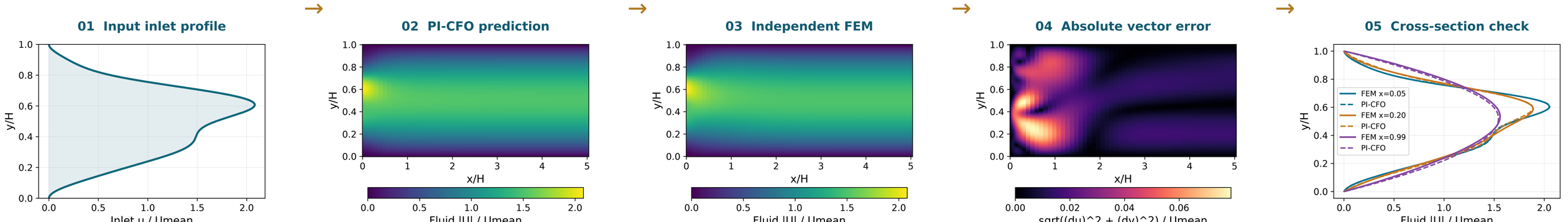
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

PI-CFO | Three-lobed inlet | Re=20 | velocity L2: 2.928% | pressure L2: 2.206%



Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

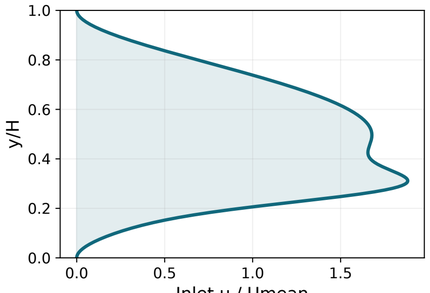
PI-CFO | Asymmetric multi-scale inlet | Re=30 | velocity L2: 2.328% | pressure L2: 5.160%



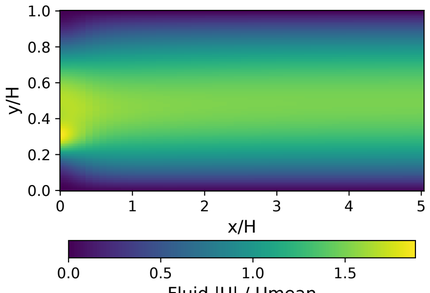
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

PI-CFO | Off-centre shoulder | Re=30 | velocity L2: 2.237% | pressure L2: 5.658%

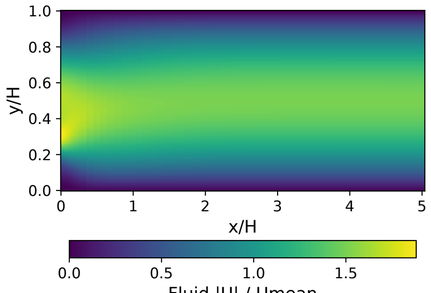
01 Input inlet profile



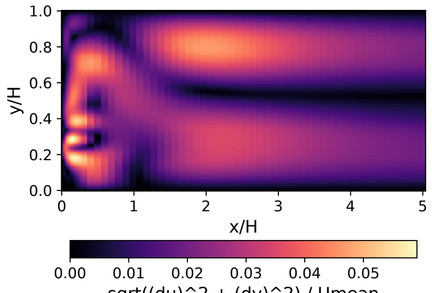
02 PI-CFO prediction



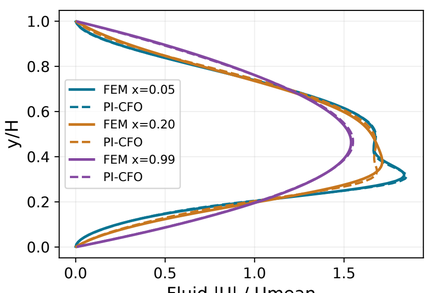
03 Independent FEM



04 Absolute vector error



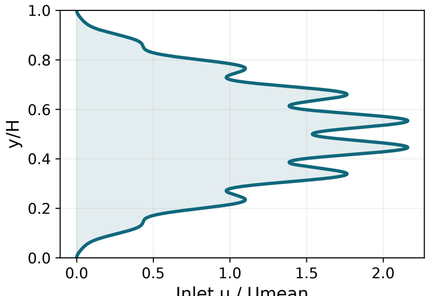
05 Cross-section check



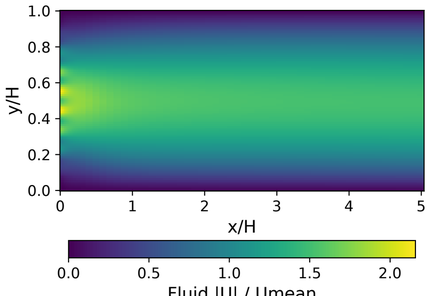
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

PI-CFO | Nine-cycle modulation | Re=55 | velocity L2: 2.753% | pressure L2: 6.839%

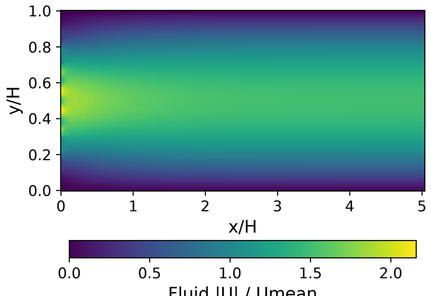
01 Input inlet profile



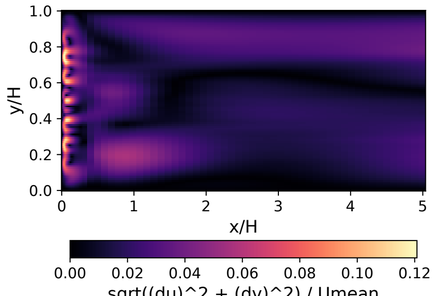
02 PI-CFO prediction



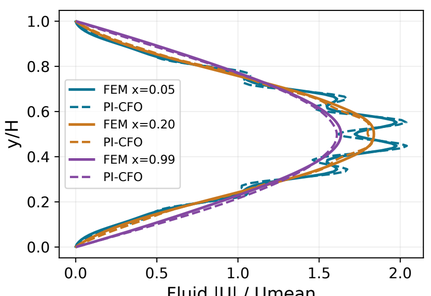
03 Independent FEM



04 Absolute vector error

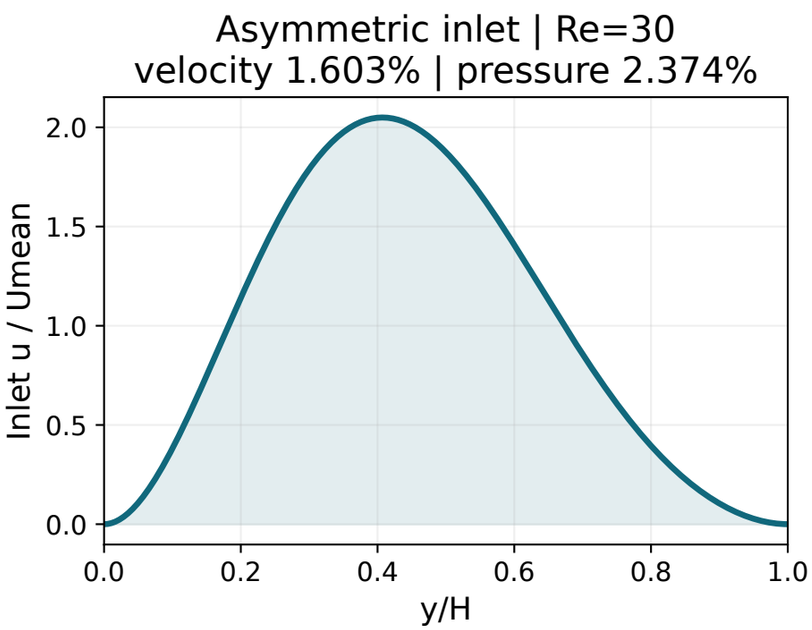


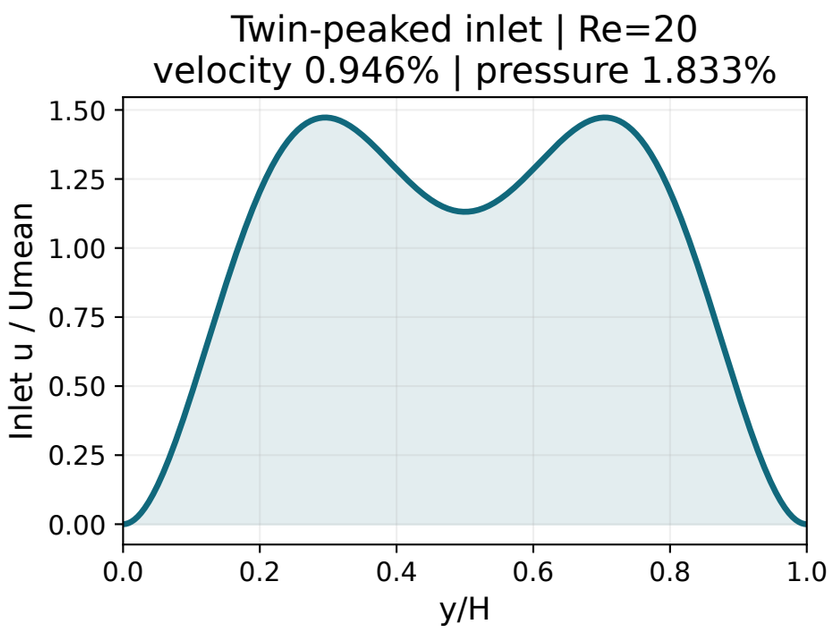
05 Cross-section check

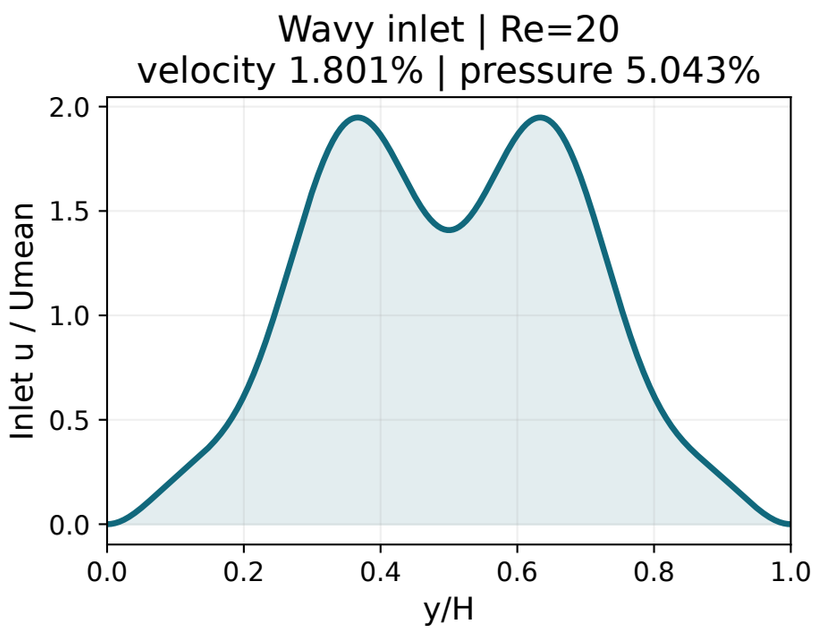


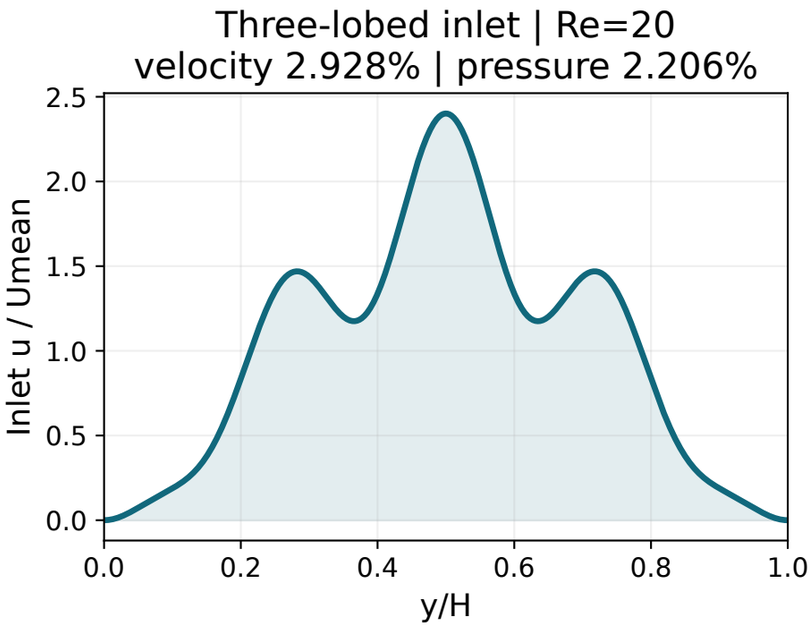
Same frozen 94k checkpoint. Independent mesh-refined FEM. Shared flow color limits. Selected exploratory examples; metrics use the saved entrance-dense grid.

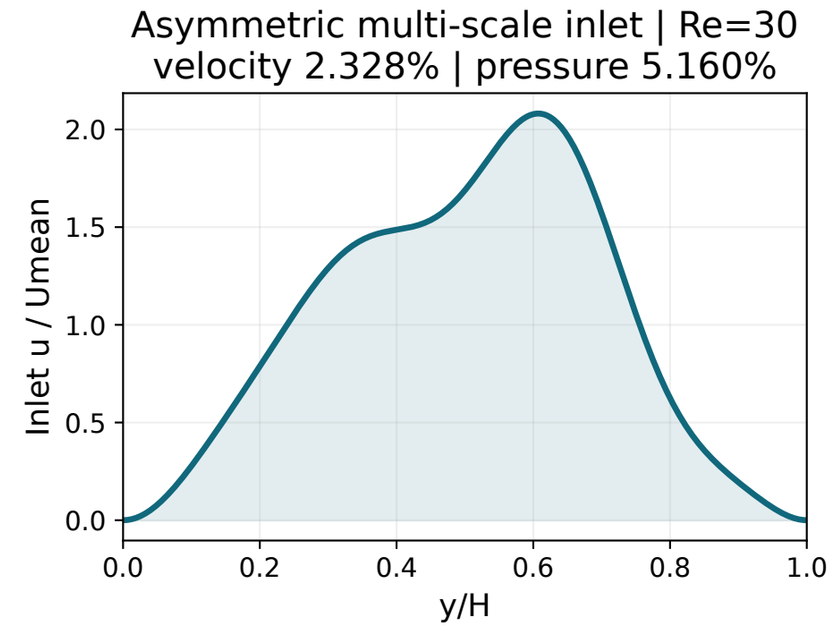
PI-CFO | Diverse inlet profiles with measured low field errors

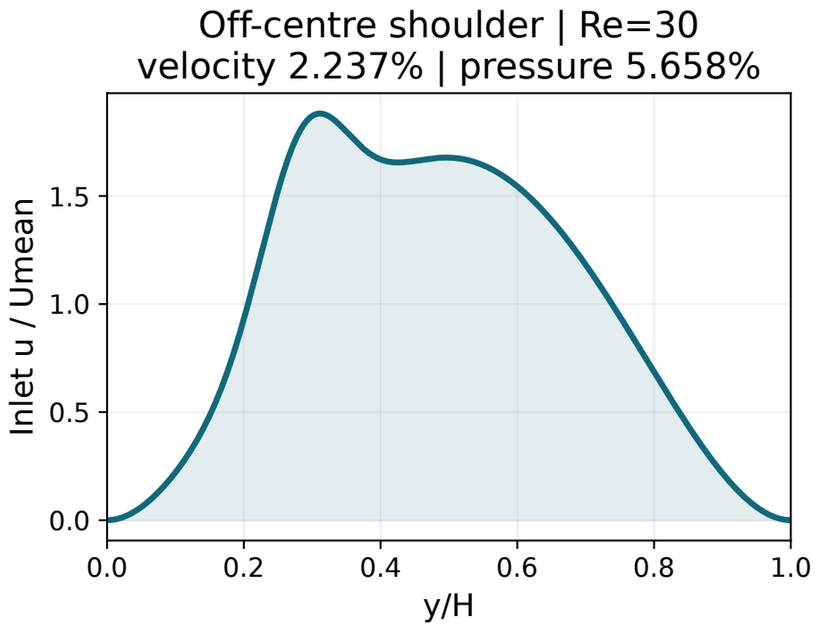


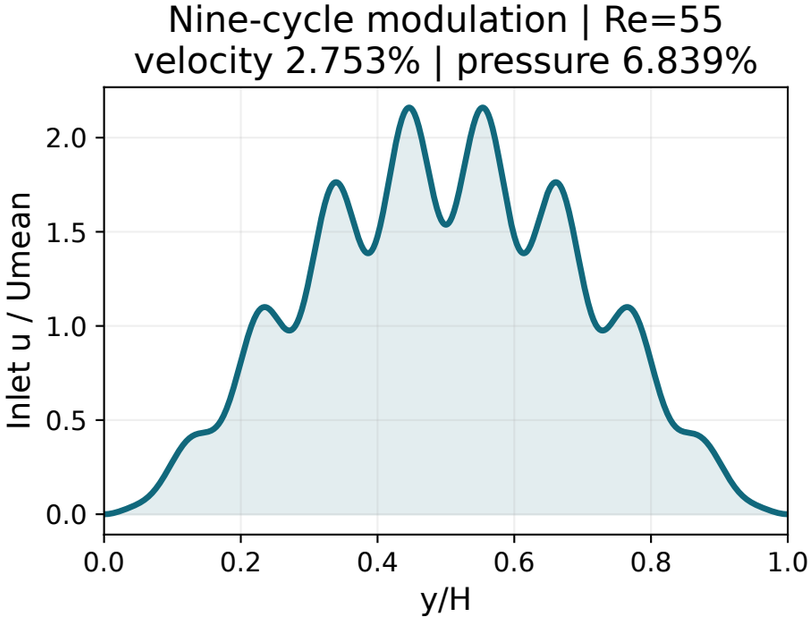






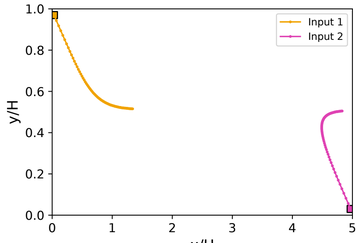




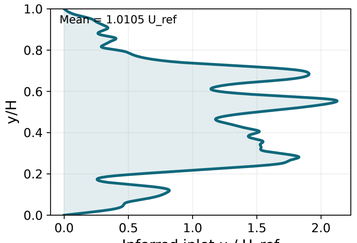


PI-CFO | Oblique launches | inferred Re=30.314

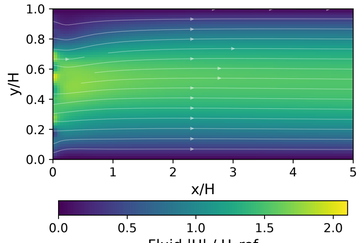
01 Given particle paths



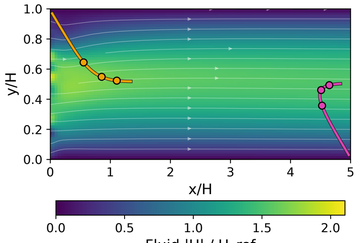
02 Inferred inlet profile



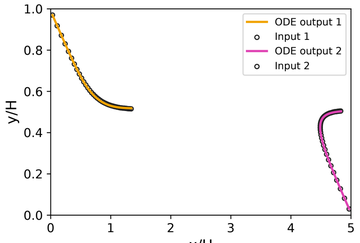
03 Inferred carrier flow



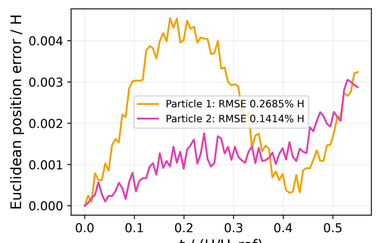
04 Particles in that flow



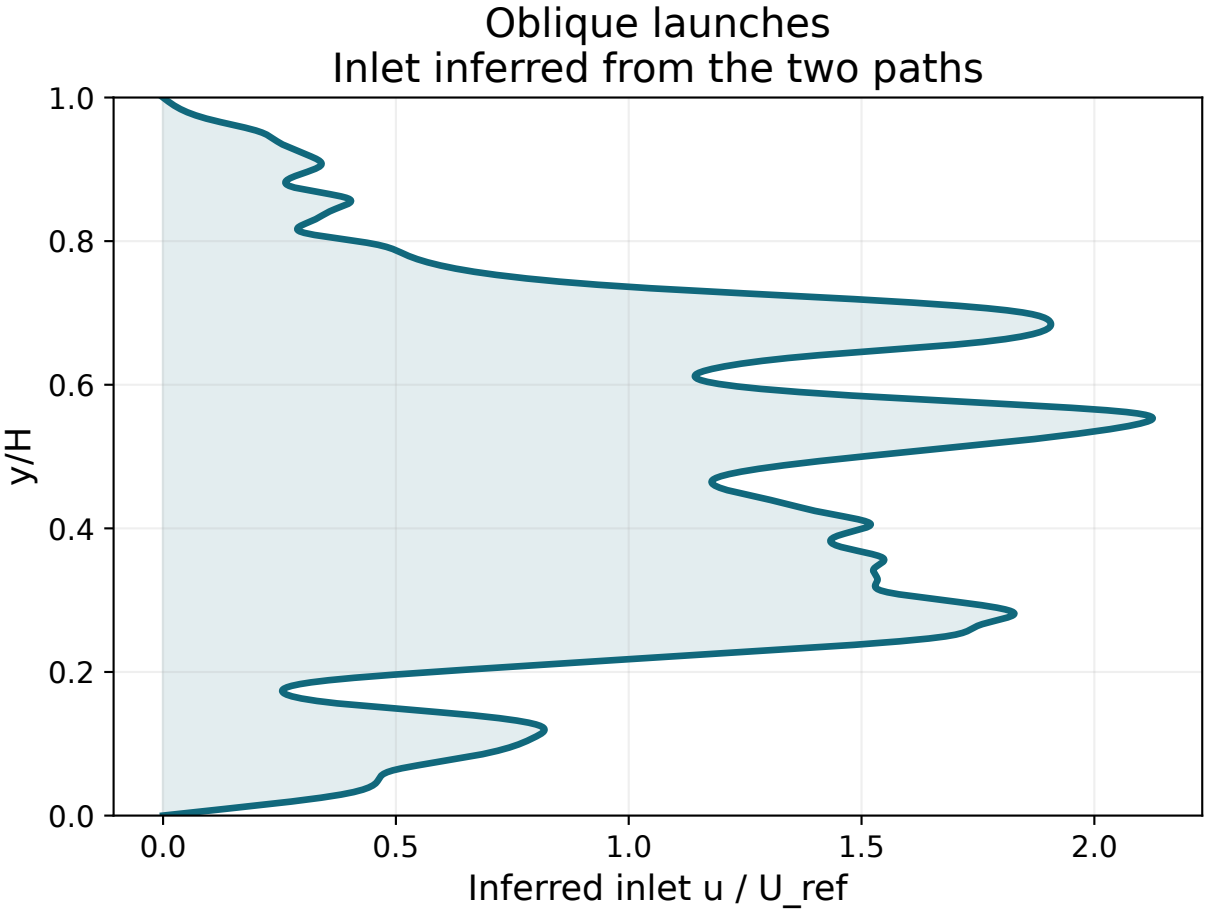
05 Input / output overlay



06 Time-aligned mismatch

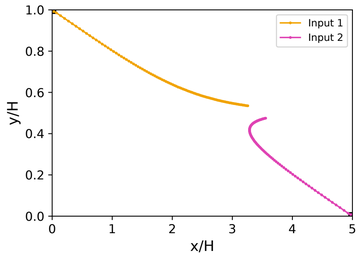


Panel 02 is the inferred inlet, not the hidden generating profile. Open circles: input paths; lines: independent particle ODE. Path agreement does not establish unique or globally accurate flow.

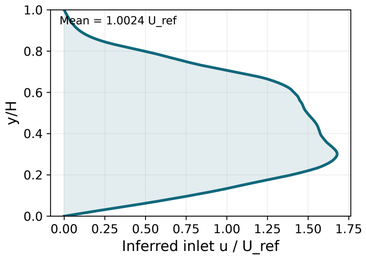

Oblique launches
Inlet inferred from the two paths
y/H
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
Inferred inlet u / U_ref

PI-CFO | Corner launches / experiment A | inferred Re=30.071

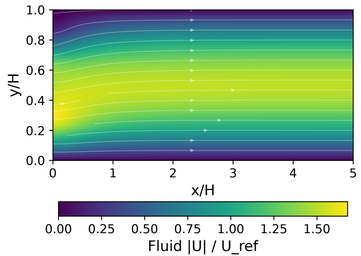
01 Given particle paths



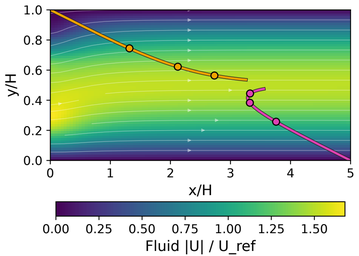
02 Inferred inlet profile



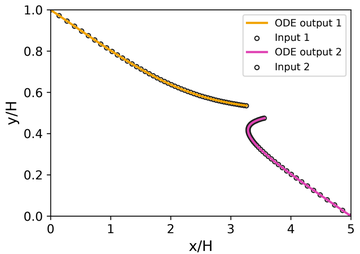
03 Inferred carrier flow



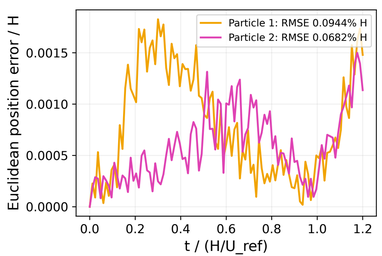
04 Particles in that flow



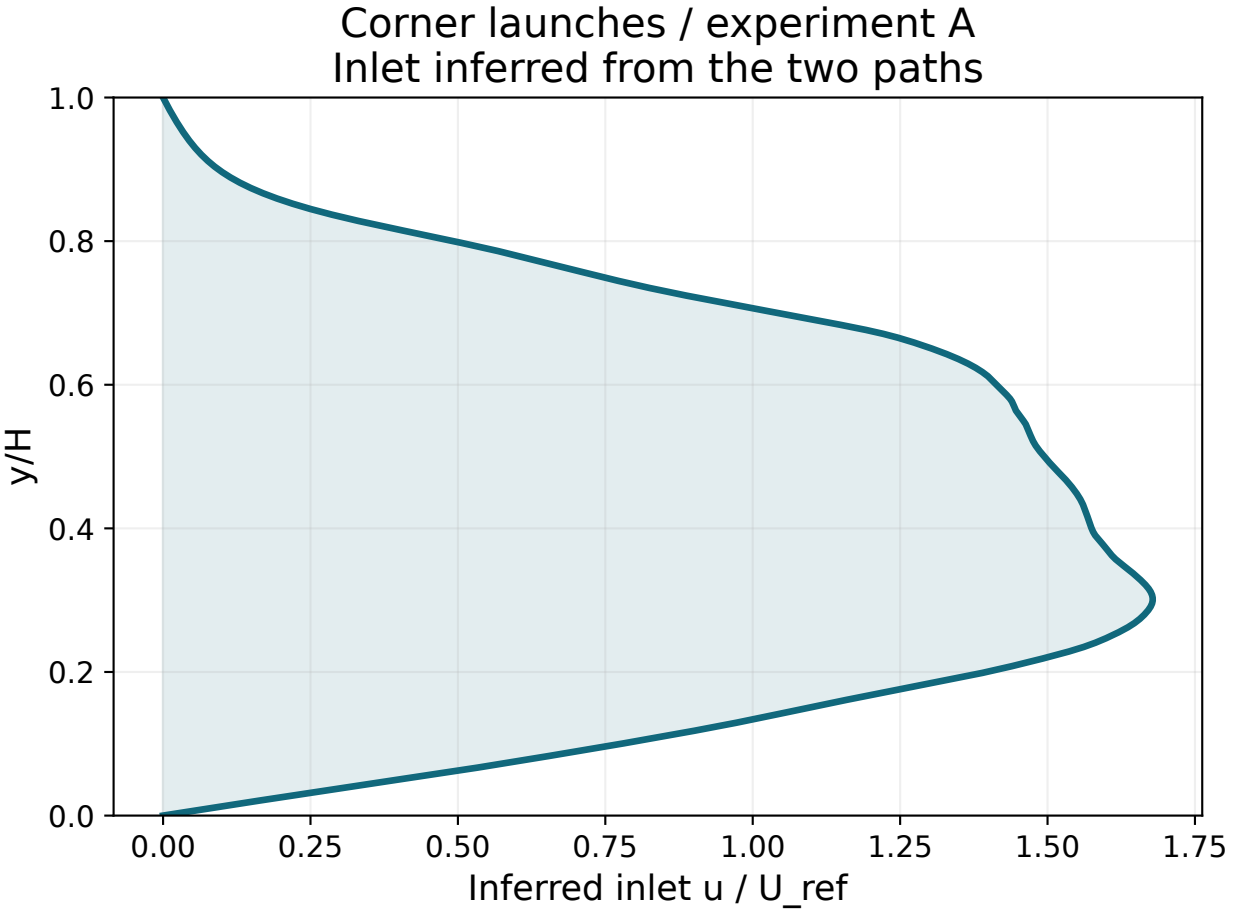
05 Input / output overlay



06 Time-aligned mismatch

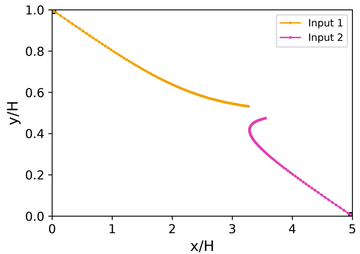


Panel 02 is the inferred inlet, not the hidden generating profile. Open circles: input paths; lines: independent particle ODE. Path agreement does not establish unique or globally accurate flow.

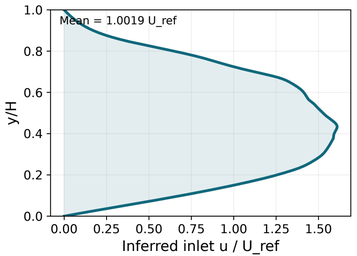

Corner launches / experiment A
Inlet inferred from the two paths
y/H
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Inferred inlet u / U_ref

PI-CFO | Corner launches / experiment B | inferred Re=30.058

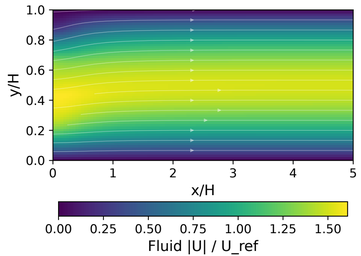
01 Given particle paths



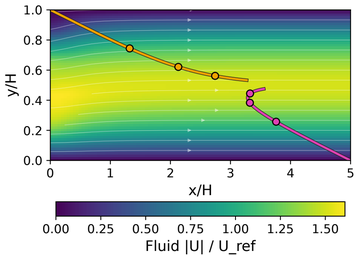
02 Inferred inlet profile



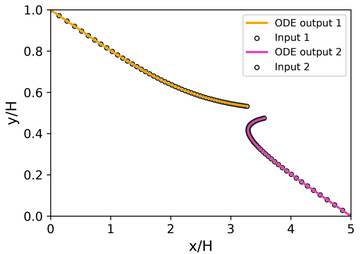
03 Inferred carrier flow



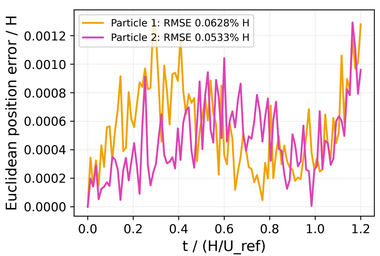
04 Particles in that flow



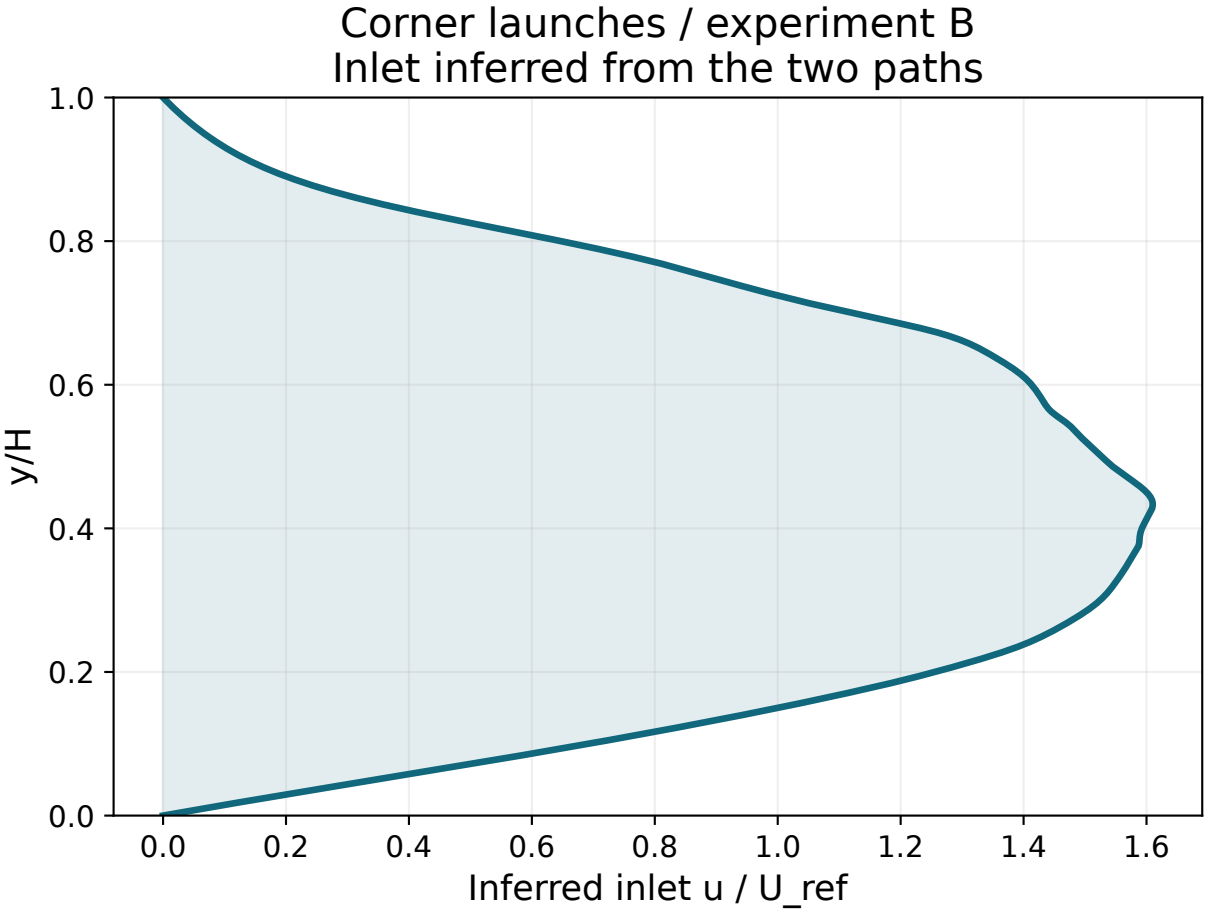
05 Input / output overlay



06 Time-aligned mismatch



Panel 02 is the inferred inlet, not the hidden generating profile. Open circles: input paths; lines: independent particle ODE. Path agreement does not establish unique or globally accurate flow.


Corner launches / experiment B
Inlet inferred from the two paths
y/H
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Inferred inlet u / U_ref